

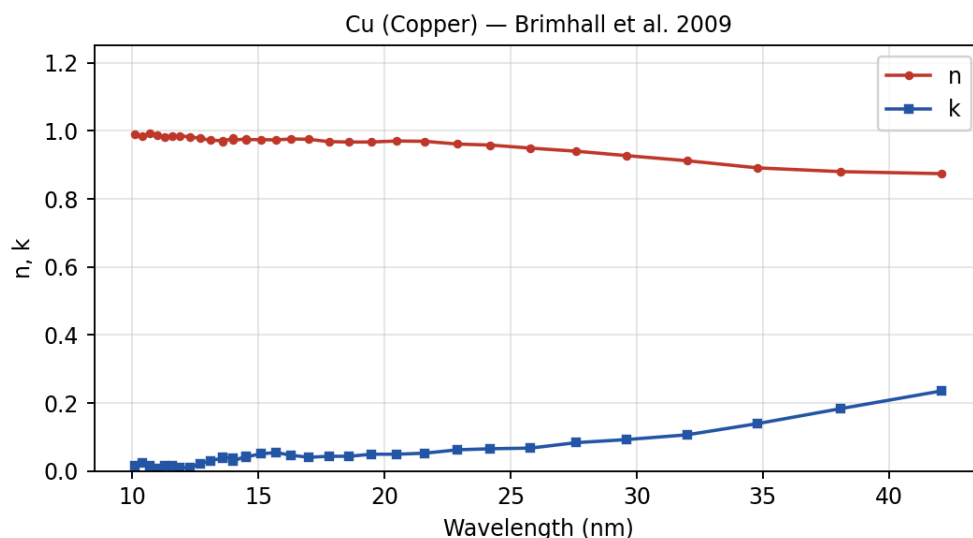
Optical constants of Cu (Copper)

Brimhall et al. 2009: n, k 0.0101–0.0421 μm

Source: <https://refractiveindex.info/?shelf=main&book=Cu&page=Brimhall> · Database record: database/data/main/Cu/nk/Brimhall.yml (public domain, CC0 1.0)

주의: 이 데이터셋의 파장 범위는 10.1–42.1 nm(극자외선, EUV)로, 가시광·근적외선(예: 532 nm, 658 nm, 1064 nm)은 포함하지 않음.

Complex refractive index ($n + ik$)



[재현 그림] 데이터베이스 레코드의 표 값으로 다시 그린 n, k 곡선 (원 페이지의 대화형 그래프에 해당).

페이지 기본 표시값 ($\lambda = 0.0421 \mu\text{m}$): $n = 0.87400$, $k = 0.23600$ · $\epsilon_1 = 0.70818$, $\epsilon_2 = 0.41253$ · $\alpha = 7.0443 \times 10^5 \text{ cm}^{-1}$ · $\text{dn}/\text{d}\lambda = -1.5000 \mu\text{m}^{-1}$ · group index $n_g = 0.93715$ · 수직 입사 반사율 $R = 0.020062$ · Brewster 각 $\theta_B = 42.155^\circ$.

References

N. Brimhall, N. Herrick, D. D. Allred, R. S. Turley, M. Ware, J. Peatross. Measured optical constants of copper from 10 nm to 35 nm. *Opt. Express* **17**, 23873–23879 (2009). <https://doi.org/10.1364/OE.17.023873>

Data (tabulated n, k)

λ (μm)	n	k	$\alpha = 4\pi k/\lambda$ (cm^{-1})	λ (μm)	n	k	$\alpha = 4\pi k/\lambda$ (cm^{-1})
0.0101	0.991	0.017	2.115e+05	0.0157	0.973	0.055	4.402e+05
0.0104	0.985	0.025	3.021e+05	0.0163	0.976	0.047	3.623e+05
0.0107	0.992	0.017	1.997e+05	0.0170	0.975	0.041	3.031e+05
0.0110	0.987	0.008	9.139e+04	0.0178	0.968	0.044	3.106e+05

0.0113	0.982	0.016	1.779e+05	0.0186	0.967	0.044	2.973e+05
0.0116	0.983	0.017	1.842e+05	0.0195	0.967	0.050	3.222e+05
0.0119	0.985	0.010	1.056e+05	0.0205	0.970	0.050	3.065e+05
0.0123	0.982	0.011	1.124e+05	0.0216	0.969	0.053	3.083e+05
0.0127	0.978	0.022	2.177e+05	0.0229	0.961	0.063	3.457e+05
0.0131	0.972	0.031	2.974e+05	0.0242	0.958	0.066	3.427e+05
0.0136	0.971	0.039	3.604e+05	0.0258	0.949	0.068	3.312e+05
0.0136	0.969	0.042	3.881e+05	0.0276	0.940	0.084	3.825e+05
0.0140	0.978	0.038	3.411e+05	0.0296	0.927	0.093	3.948e+05
0.0140	0.972	0.030	2.693e+05	0.0320	0.912	0.107	4.202e+05
0.0145	0.976	0.044	3.813e+05	0.0348	0.891	0.140	5.055e+05
0.0145	0.974	0.042	3.640e+05	0.0381	0.880	0.184	6.069e+05
0.0151	0.974	0.051	4.244e+05	0.0421	0.874	0.236	7.044e+05

n, k, λ 는 데이터베이스 레코드 원문 그대로(34행; 0.0136, 0.0140, 0.0145 μm 는 원문에 값이 두 개씩 있음). α 열은 원문에 없는 계산값($\alpha = 4\pi k/\lambda$).

Cu (Copper)의 다른 데이터셋 (같은 사이트)

Page	범위
Johnson and Christy 1972	n,k 0.188–1.937 μm
McPeak et al. 2015	n,k 0.3–1.7 μm
Babar and Weaver 2015	n,k 0.2066–12.40 μm
Brimhall et al. 2009 (이 문서)	n,k 0.0101–0.0421 μm
Werner et al. 2009	n,k 0.01759–2.480 μm (실험 / DFT 계산 각각)
Stahrenberg et al. 2001	n,k 0.128–0.496 μm
Ordal et al. 1985	n,k 0.517–55.6 μm
Querry 1985	n,k 0.21–55.6 μm
Hagemann et al. 1974	n,k 0.0000248–248 μm
Rakić et al. 1998	Brendel-Bormann / Lorentz-Drude model; n,k 0.207–12.4 μm

About Copper

Copper (Cu) is a highly versatile transition metal with excellent electrical and thermal conductivity. While its primary applications span electrical wiring, plumbing, and various industrial machinery, it also finds specialized uses in optics. Notably, polished, bare copper is often the material of choice for high-power infrared mirrors, including those used in CO₂ lasers, due to its high reflectivity in the infrared spectrum and exceptional thermal resilience. The metal's malleability and corrosion resistance further allow it to be shaped into intricate forms, making it invaluable across a range of industries. Whether in general construction or specific niches like high-power optical components, copper's multifaceted properties make it a material of critical importance.